

PROJECT SYNOPSIS

Personal Agentic RAG

One private AI assistant for your studies, resume, GitHub, certificates and deadlines.

“Handle your documents safely and securely.”

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THE PROBLEM

Everything about you is scattered



Studies

Notes, books and past papers in ten folders



Deadlines

Exams and applications tracked in your head



Documents

Certificates lost across drives and phones



GitHub

Great projects that never reach the resume



LinkedIn

Profile falls out of date with real work



Resume

Many versions, edited by hand for every job

Update one thing, and you have to update everything **by hand**.

THE IDEA

One assistant that knows you, and only you



Study

“Explain deadlock from my OS notes” answers with the page number.



Ask

“What was my CGPA in semester 5?” gets an answer plus the source document.



Fetch

“Give me my AWS certificate” finds the file in your encrypted vault.



Track

Upcoming exams, certification expiries and application deadlines.



Tailor

Paste a job description to get a match score and a tailored resume draft.

The AI **drafts**, and **you approve**. It never submits or posts on its own.

A study companion built on your own notes



Your study library

- Lecture notes, typed or handwritten
- Textbooks and reference PDFs
- Lecture slides
- Previous year question papers
- Syllabus for each course

Organised as Semester → Course → Topic

PHASE 1

Ask with page citations

“Explain deadlock” answers from *your* OS notes, page 12

PHASE 2

Summaries and flashcards

For any unit, chapter or whole course

PHASE 2

Practice from past papers

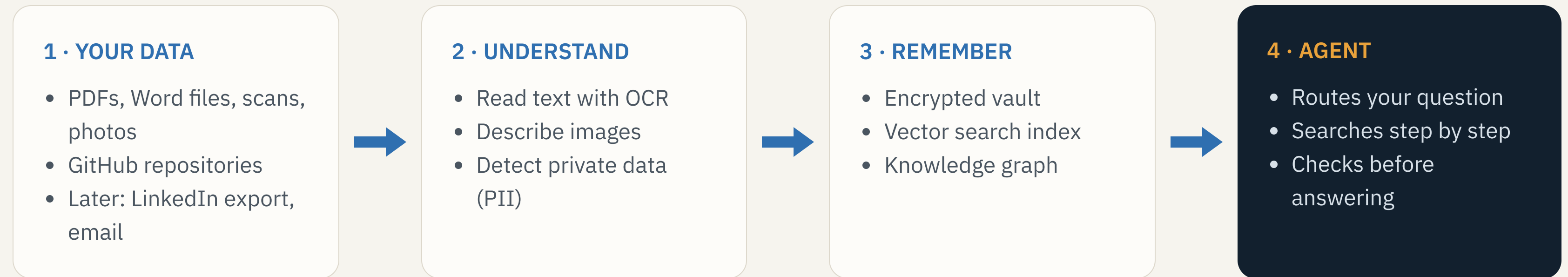
Finds frequently asked topics and builds quizzes

PHASE 2

Study planner

Splits the syllabus into daily topics before each exam

How it works



Open-source AI models run on **your own machine**, so your documents never go to an AI company.

What happens when you ask a question

1

Question

You type in the web app

2

Router

A fast decision model
picks the path and
blocks unsafe requests

3

Agent

Searches documents,
graph and GitHub,
several times if needed

4

Check

Is the answer
supported by the
sources? If not, it retries

5

Answer

With citations, or an
honest “I don’t know”

You: Which of my projects used FastAPI?

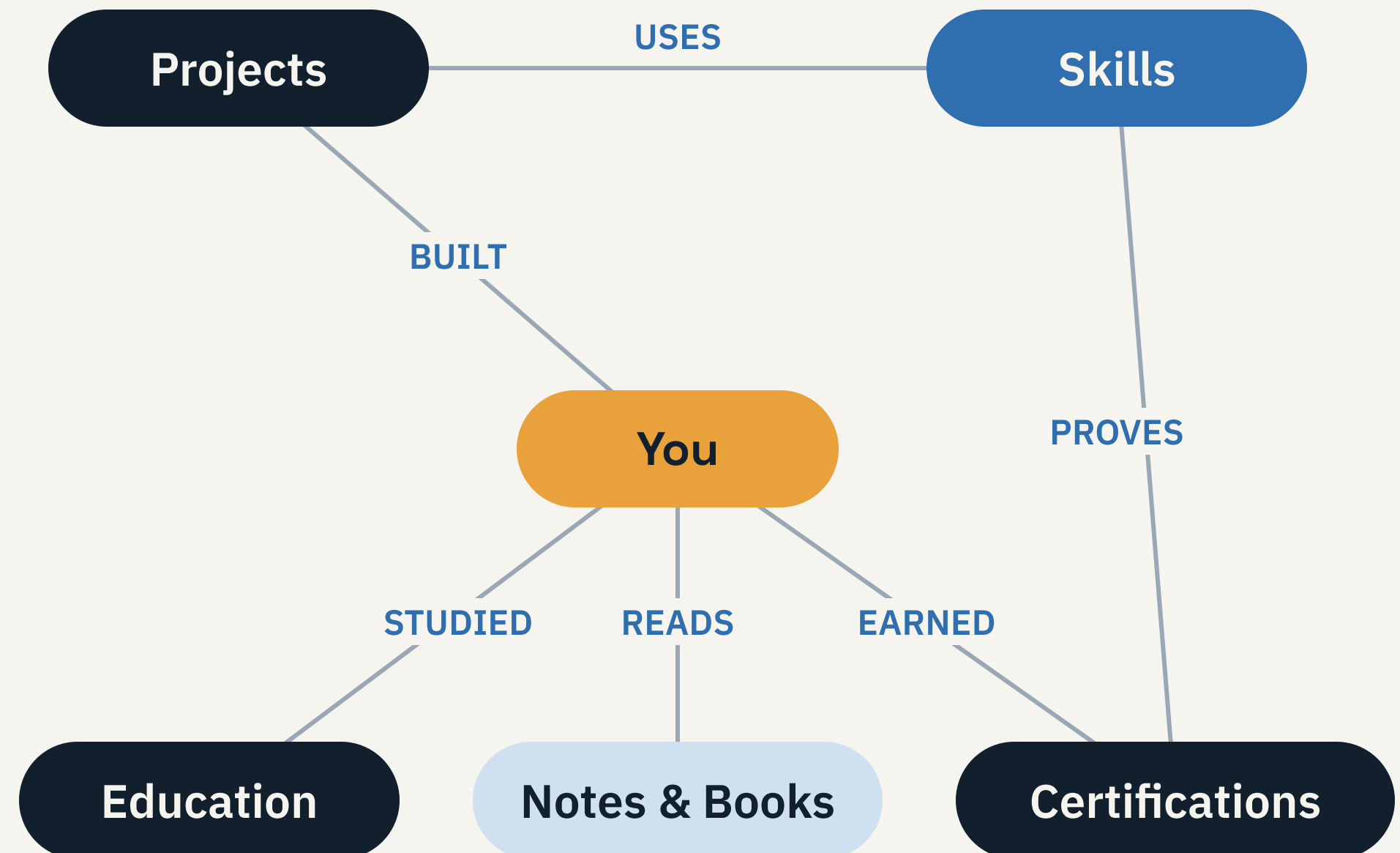
Assistant: “Campus Events Portal” and “Expense Tracker API”. Sources: their GitHub READMEs.

A map of you

Plain search finds matching text. A knowledge graph also knows how things are **connected**.

EXAMPLE QUESTION

“Which skills from my DBMS course have I used in a project?”



SECURITY BY DESIGN

Six layers protect your documents



Private network

Not reachable from the public internet



Passkey login

Fingerprint or Face ID, with no passwords to leak



Encrypted vault

Every file is encrypted at rest (AES-256)



Local AI models

No data is sent to AI companies



Guardrails

Blocks prompt injection hidden in files



Audit log

Every access and action is recorded

BUILT FOR A STUDENT BUDGET

100% open source, zero license cost

O

paid software or API fees. It runs on a normal laptop during development.

Part	Open-source tool	Cost
AI models	Ollama + small open models	Free
Reading documents	Docling + OCR	Free
Agent	LangGraph	Free
Database + search	PostgreSQL + pgvector	Free
Monitoring	Langfuse (self-hosted)	Free
Private access	Tailscale free plan	Free
Cloud (later)	Student credits	Optional

Built in modules, one phase at a time

PHASE 1 · FOUNDATION

Vault, Study + Ask

- Upload and encrypt documents
- Study library for notes and books
- OCR for scans and photos
- GitHub connector
- Chat with citations
- Simple knowledge graph

PHASE 2 · CAREER

Career + exams

- Flashcards, quizzes, study planner
- Deadlines and exam tracker
- Job description match score
- Auto-tailored resume draft
- Full Graph RAG
- MCP server

PHASE 3 · AUTOMATION

Hands

- Fill job forms in the browser
- LinkedIn post and profile drafts
- Email and calendar sync
- Mobile app
- Cloud deployment

WHY IT’S DIFFERENT

Compared with existing tools

Feature	Cloud AI chatbots	Resume tools	This project
Data stays on your machine	No	No	Yes
Knows all your documents and GitHub	Partly	No	Yes
Answers cite their sources	Partly	No	Yes
Study help from your own notes, with page numbers	Partly	No	Yes
Job match and tailored resume	Manual	Yes	Yes
Open source and free	No	Mostly no	Yes
Acts only with your approval	Varies	Varies	Always

How we will measure success

Answer accuracy

50 test questions about a fake demo student and open textbooks, with known answers

Faithfulness

How often answers are backed by the retrieved sources, measured with Ragas

Security tests

Planted prompt-injection documents must be ignored, and private data never leaks

Speed and cost

Response time on a student laptop, and a cost that stays at zero

Thank you

Questions and feedback welcome

Open source from day one: github.com/MohitBareja16/personal-agentic-rag